



Teaching Machines to Read Signals: AI Meets Ethical Hacking

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Hello everyone, my project focused on Radio Frequencies and using Artificial Intelligence to help demodulate and better receive a transmission.



Presentation Overview

- Why Signals Matter In Ethical Hacking
- Traditional Network Layers Vs. RF Signal Layers
- Dataset and AI Model Overview
- Feature Engineering and Feature Selection
- Simulation: Modulation and Encryption Layers
- Key Takeaways

As an overarching summation of my project, I'll go over why signals matter in ethical hacking, the similarities of the 5 layers in ethical hacking and listening to RF signals, briefly go over the AI pipeline since more of the focus is the ethical hacking concepts, a simulation of modulation and encryption of a signal to help better understand the narrative, and some key takeaways from my project.



Why Signals Matter In Ethical Hacking

So. Why did I think signals would matter or at the very least, align with ethical hacking concepts we learned.

What We Learned Vs. Signals Application

WHAT WE LEARNED

- Reconnaissance (PASSIVE :D)
- Enumeration
- Gaining Access
- For servers, databases, and web applications



Much of what we studied in ethical hacking focused on gaining access to systems such as servers, databases, and web applications. Whether it be through passive reconnaissance and/or enumeration in order to gain access to our PTG flags. Or at least an image showing we did the thing we were supposed to do and if nothing else, we gave the good 'ol college try!

What We Learned Vs. Signals Application

SIGNALS

- Information travels over wavelengths (Signals)
- Signals carry encoded information (Bitstream)
- Transmitter alters bitstream
- If Receiver knows alterations, bitstream is received and information is understandable



But before information reaches those systems, it often travels across physical communication channels: wired networks, Wi-Fi, cellular, Bluetooth, satellite, or radio-frequency signals. These signals carry encoded information. In theory, if an authorized analyst can capture a signal, identify its modulation, recover the bitstream, and handle any encryption correctly, the transmitted information can be interpreted.

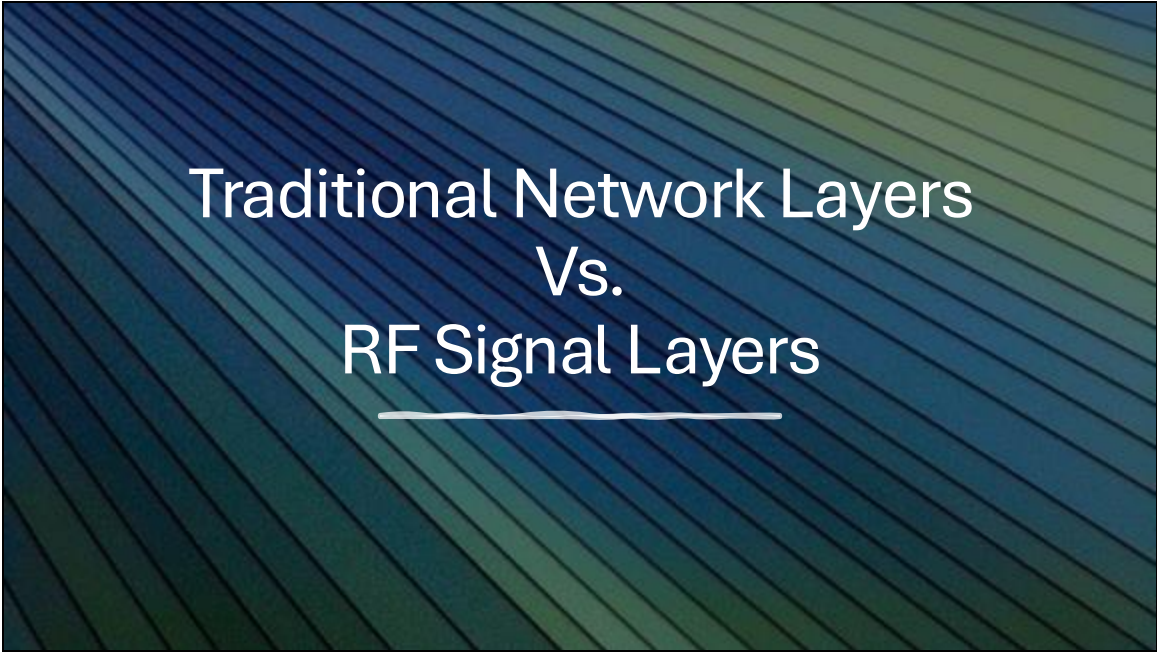
What We Learned Vs. Signals Application

THE BRIDGE

- We practiced with information that arrived or already existed within a machine
- Signals allows us to see information during transit, but can be protected in layers like what we learned



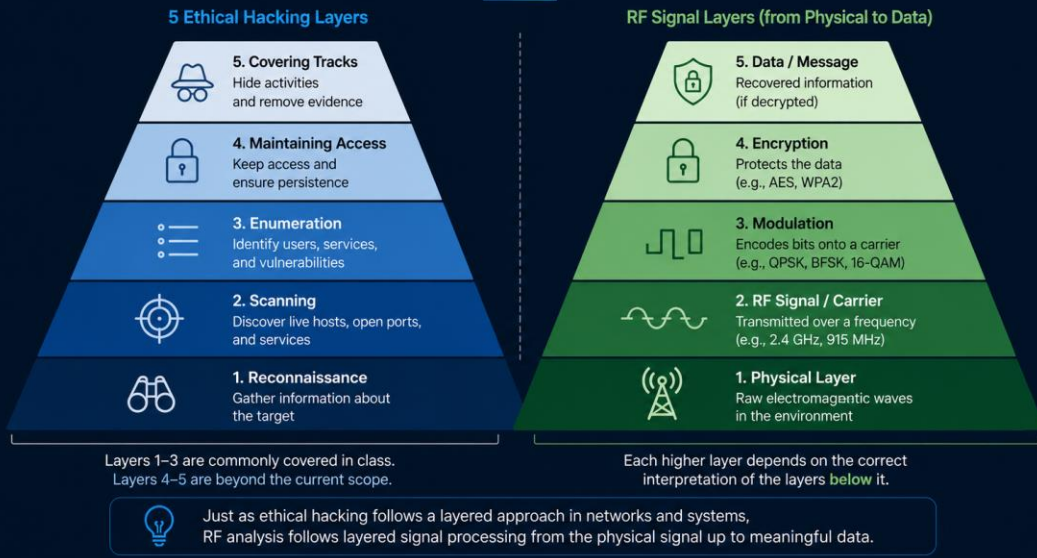
In other words, we mostly looked at systems after the data had already reached a server, database, or web application. RF security looks one layer lower. Before data becomes a login request, packet, or database query, it may have traveled as a physical signal. If we can understand that signal's structure, we can begin to interpret the information it carries — but encryption and protocol layers still protect the actual content.



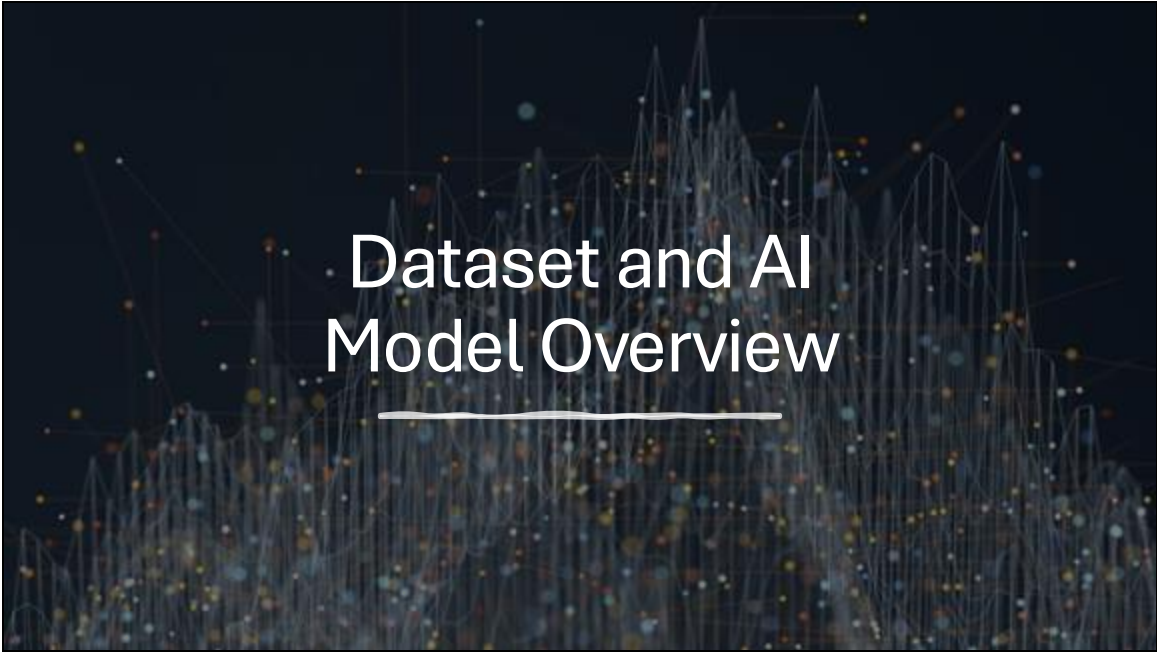
Traditional Network Layers Vs. RF Signal Layers

Key word in that explanation, layers.

Traditional Network Layers vs. RF Signal Layers



From my understanding of the 5 layers we were taught in class – the ethical hacking layers on the left – we only went from layers 1-3 for obvious ethical reasons. In general, we used critical thinking to better understand where we could PASSIVELY learn more about our target, used things like Wireshark and Netcat for port scanning, and banner grabbing to identify any vulnerable services and their versioning. Similarly with RF signals, we need to position a radio receiver antennae where we believe our desired signals exist, accurately capture the signals in the area and scan for the signal we want out of potentially hundreds or more signals captured, and once we have that filtered down signal, we need to first understand the signal's modulation type before being able to do any sort of further exploitation, just like banner grabbing.

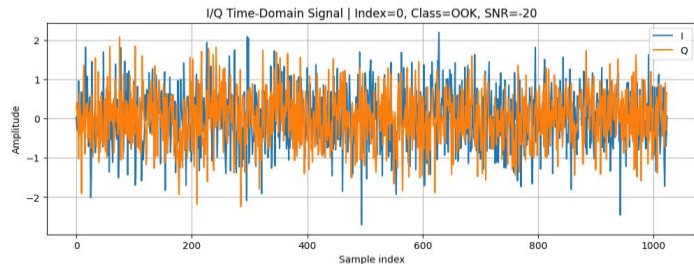
The background of the slide is a dark, deep blue or black. It is filled with a complex network of thin, light blue or grey lines that connect various points. These points are represented by small, glowing spheres in shades of blue, yellow, and orange. The overall effect is reminiscent of a data network, a molecular structure, or a complex signal waveform. The lines and nodes are more densely packed in some areas, creating a sense of depth and complexity.

Dataset and AI Model Overview

This brings me to the dataset I used and the AI model I trained for detecting a signal's modulation type.

Full Dataset

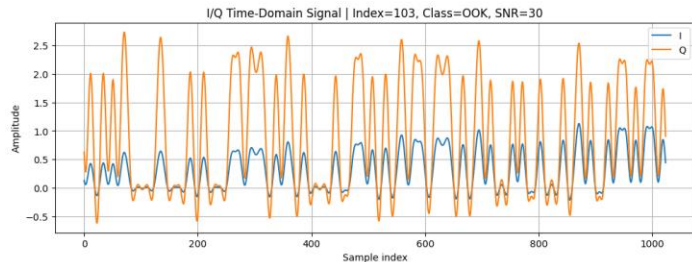
- [DeepSig's RADIOML 2018.01A](#)
- 2,555,904 Signals
- 24 Modulation Types
 - Uniform Distribution
- 3 Features
 - I/Q (1024 array * 2)
 - Modulation ID
 - Signal-to-Noise Ratio (SNR: [-20, 30])



Thankfully, DeepSig already had a very large and reputable dataset with signals ranging from very noisy to very clean. This massive dataset captured the full I/Q of each of the 2.5 million signals where each signal was one of 24 modulation types and uniformly distributed and these signals could be observed at a very low signal to noise ratio or a very high SNR across the entire dataset. The image is an OOK modulation type signal with the lowest SNR at -20 . So a very noisy signal.

Reduced Dataset

- [DeepSig's RADIOML 2018.01A](#)
- 24,000 Signals
- 24 Modulation Types
 - Uniform Distribution
- 3 Features
 - I/Q (1024 array * 2)
 - Modulation ID
 - Signal-to-Noise Ratio (SNR=30)



For my purposes and the initial "proof-of-concept" so to speak, I wanted to first work with the cleanest possible signals and definitely not because my tiny laptop couldn't handle anything more! This made things much easier to process for my models reducing the dataset to 24000 signals, still uniformly distributed, but only signals at the highest SNR, 30. The image is another OOK modulation type signal, but as you can see, an extremely clean signal capture.

Random Forest Classifier

An ensemble of decision trees that improves accuracy and reduces overfitting by averaging the predictions of multiple trees.

1 Bootstrap Sampling

Randomly sample (with replacement) from the training data to create multiple datasets.

2 Train Multiple Trees

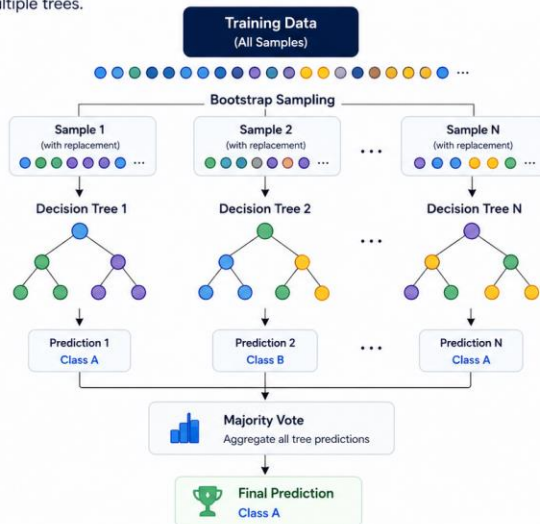
Train a decision tree on each bootstrap sample. Each tree sees a different subset of the data.

3 Aggregate Predictions

Each tree makes a prediction. The final output is determined by majority vote (classification).

4 Final Prediction

The class with the most votes is the Random Forest prediction.



Why Random Forest?



High Accuracy
Combines many trees to reduce error.



Robust to Overfitting
Averaging reduces variance compared to a single tree.



Handles Large Datasets
Works well with high-dimensional data and missing values.



Feature Importance
Provides insight into which features matter most.

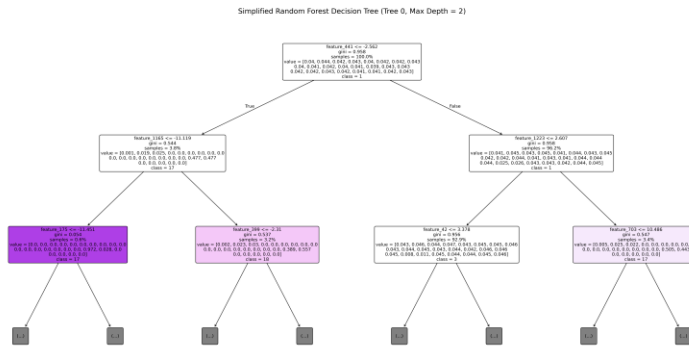
Key Points

- Uses bagging (bootstrap aggregating)
- Each tree is trained on a random subset
- Predictions are combined by majority vote
- Reduces overfitting and improves generalization

As for the AI model, I chose a Random Forest Classifier – and for brevity's sake – a Random Forest Classifier is just a decision tree where multiple "people" go through the decision tree and take a majority vote on what the answer is. In other words, I wanted multiple "people" to look at this data and take a vote on what a given signal's modulation type was.

Baseline Model

- Input:
 - 2,048 Features
 - 24,000 Signals
 - 24 Modulation Types
 - Uniform Distribution
- Output:
 - 1of24 Modulation Types
- Performance:
 - Accuracy ~36.21%



As a baseline, sending the reduced dataset as is, the overall accuracy of the AI model correctly identified approximately 36.21% of a the signal's modulation type out of 24. Honestly, I was surprised at how high it performed, but then again, having the cleanest possible signal likely had much to do with this performance.

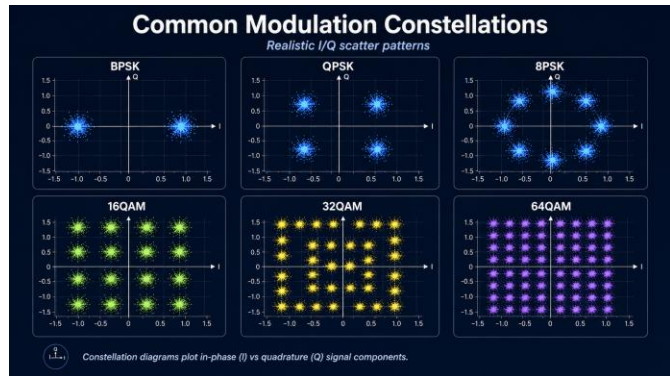


Feature Engineering and Feature Selection

Now for the more interesting thing when training any sort of AI in my personal opinion. Feature engineering and feature selection, in other words, from our reduced dataset, is there any information I can extract that should have a stronger correlation to the modulation type of a signal.

The SIGINT Pipeline

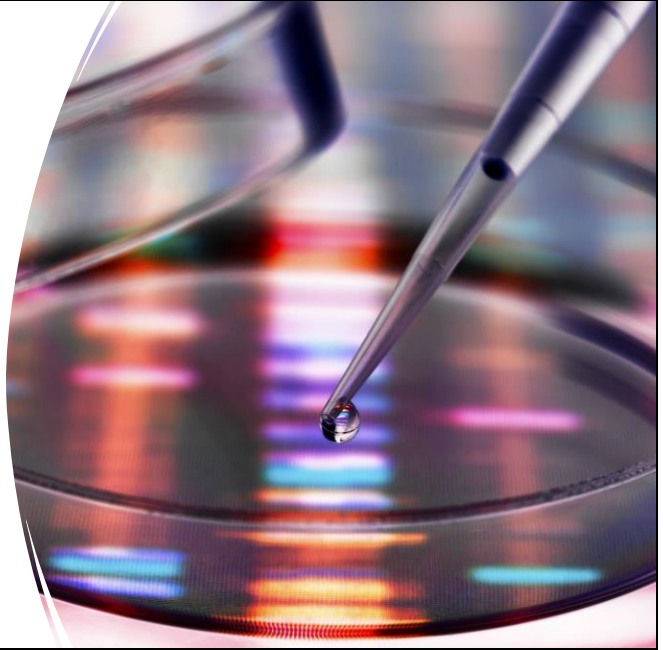
- SIGINTers apply complex mathematics to garner more information about signals like:
 - I/Q stats
 - Magnitude/Power
 - Phase/Frequency
 - FFT/Spectral Features
 - Constellations



In traditional signal analysis, an analyst applies transformations like magnitude, phase, power, or FFT analysis to better understand a signal. These are explained as helpful for identifying a signal's modulation type. I applied the same idea computationally with python functions to generate engineered signal features, then I would use these to train the Random Forest to classify modulation types.

Feature Engineered Dataset

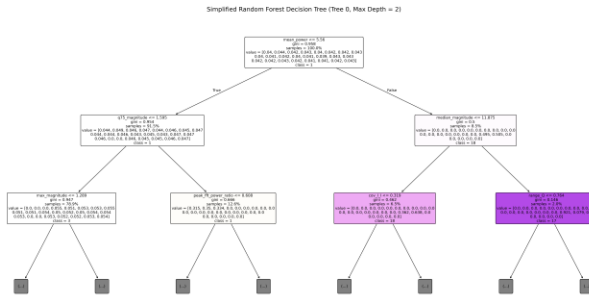
- 24,000 Signals
- 24 Modulation Types
 - Uniform Distribution
- 105 Features
 - I/Q stats
 - Magnitude/Power
 - Phase/Frequency
 - FFT/Spectral Features
 - Constellations



By doing so, I took out the raw I/Q data that when flattened, became 2048 features, and engineered 105 features that should be more information dense and more helpful when training the AI model.

Feature Engineered Model

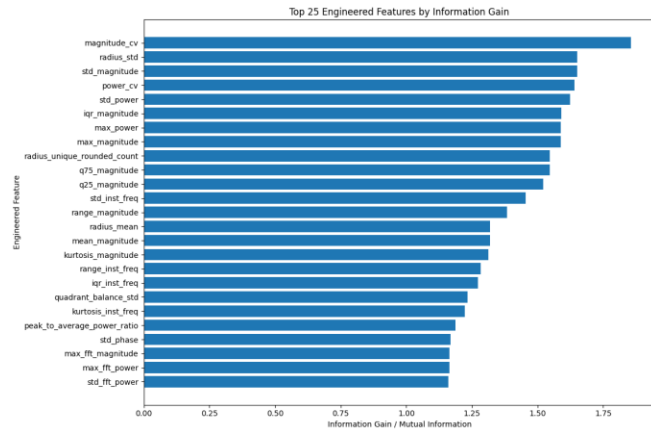
- Input:
 - 105 Features
 - 24,000 Signals
 - 24 Modulation Types
 - Uniform Distribution
- Output:
 - 1 of 24 Modulation Types
- Performance:
 - Accuracy ~72.17%



Similar to before, I sent this feature engineered dataset as is for AI model training and testing and it returned nearly double the accuracy performance at approximately 72.17%!

Feature Selection

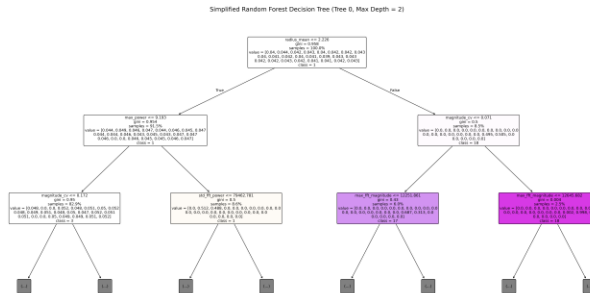
- Each feature can generate an information gain score
- Higher information gain leads to higher confidence in decisions



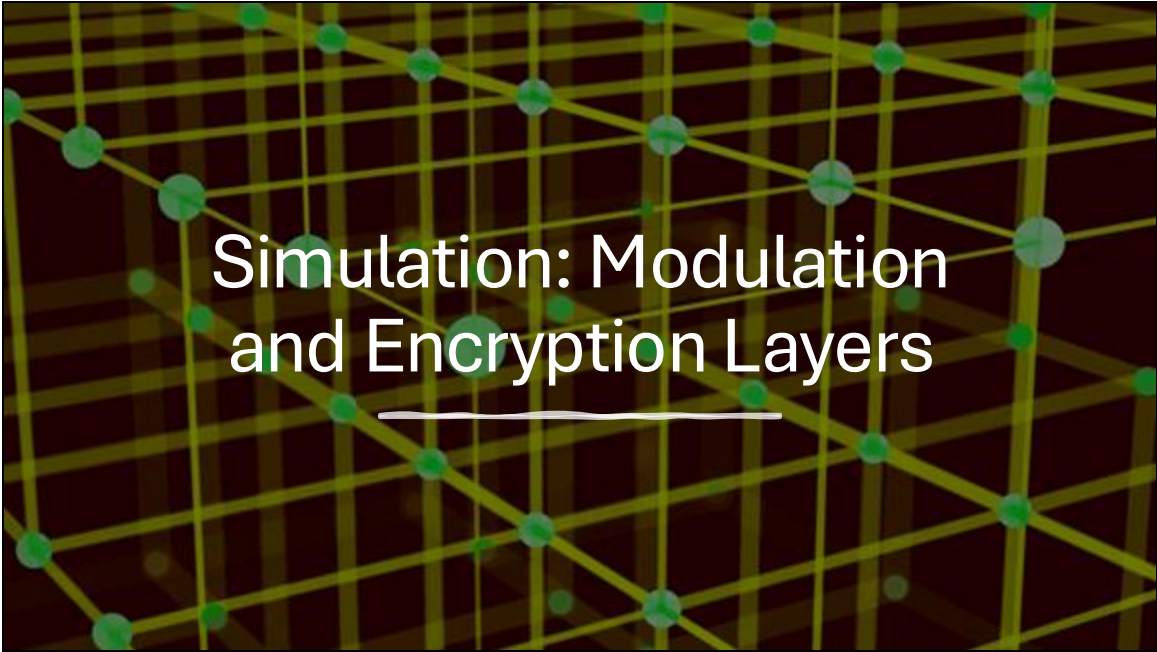
Not satisfied with arbitrarily training a model on 105 features, I went through and tied an "information gain" value to every feature and their relation to arriving at the modulation type and selected the top 25 features from this array.

Top 25 Feature Model

- Input:
 - 25 Features
 - 24,000 Signals
 - 24 Modulation Types
 - Uniform Distribution
- Output:
 - 1 of 24 Modulation Types
- Performance:
 - Accuracy ~69.37%



Training and testing the AI model on the top 25 most informative features had a slight drop in accuracy performance at approximately 69.37%, but the key point here being that I needed less than ¼ of the features I engineered to have a model that maintained over 96% of the accuracy performance. In terms of scalability and resources, this is extremely significant!

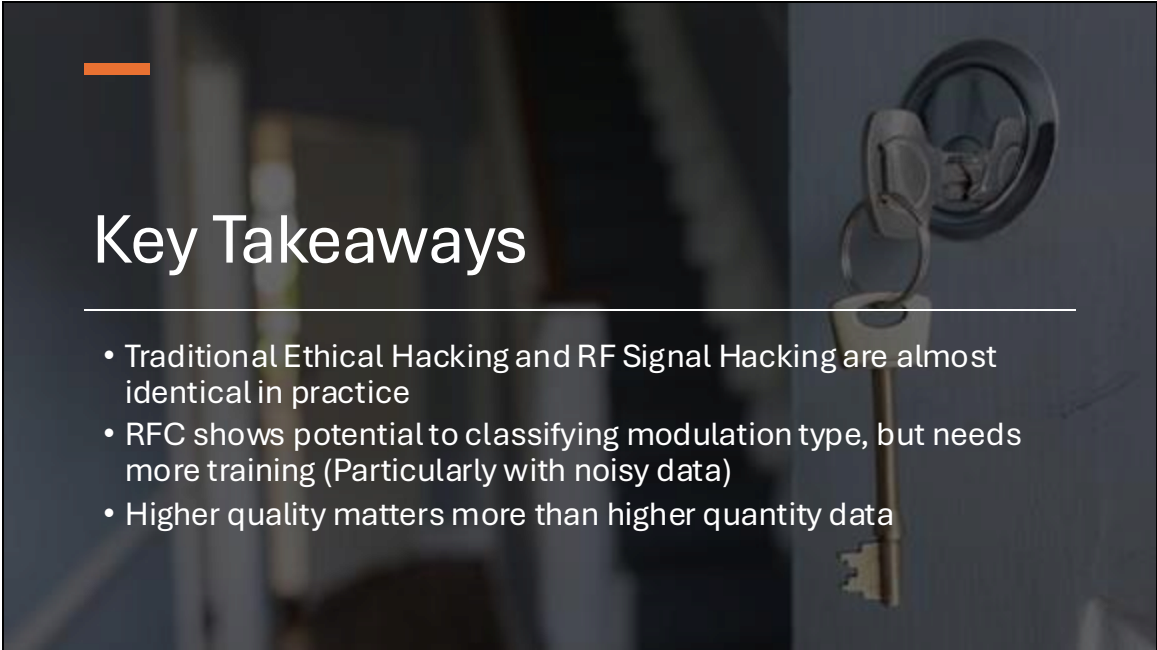


Simulation: Modulation and Encryption Layers

So at the very least, I have an AI model that can accurately guess a signal's modulation about 70% of the time assuming the signal is one of the 24 modulation types and is very clean. Just like banner grabbing with ncat, this AI model is doing a similar process. As for the encryption portion, I created a simulation since in theory, an encryption layer for a signal could be 2^{256} bits long which would be impossible to brute force with any known system! Also, it's not exactly legal for me to just record encrypted signals and try to decrypt them! So simulations are the next best thing and should help better understand the overall ethical hacking concept as it relates to RF signals!

The figure compares two audio reconstruction results. On the left, under the heading 'Wrong Mod/Encryption', there is a speaker icon inside a light gray circle, representing a poor or distorted audio output. On the right, under the heading 'Correct Mod/Encryption', there is a similar speaker icon inside a light gray circle, representing a clear and accurate audio output. The background of the entire figure is a blue and green fractal pattern.

On the bottom summarizes the output text. As another example of having the wrong modulation and encryption vs. the correct modulation and encryption, here's two audio files. One sounds very distorted and timing may be incorrect; On the other hand, the other one is clear as day!



Key Takeaways

- Traditional Ethical Hacking and RF Signal Hacking are almost identical in practice
- RFC shows potential to classifying modulation type, but needs more training (Particularly with noisy data)
- Higher quality matters more than higher quantity data

With the simulation out of the way, the key takeaways from my project is how extremely similar the traditional ethical hacking concept is to RF signal hacking, though much is left to be explore with the AI model, it shows much promise in expediting or at least augmenting a signal analyst's pipeline, and most importantly for AI training, the quality of data is a much more significant difference in AI model performance than the quantity of data is!

Resources Used

- [DeepSig's RADIOML 2018.01A](#) (Original Dataset)
- [PySDR](#) (General SDR Knowledge)
- Various Sources on SDR (Supplementary SDR Knowledge, in writeup)
- Microsoft Powerpoint AI Designer (Slide Setup)
- ChatGPT Image Creation (Slide Setup)

These are some key resources I used and the "various sources" was way too long to list here so I'm putting them in my writeup for my AI class assignment.



Thank you for your time and I hope this was an interesting and entertaining presentation! Have a great Summer everyone!